CLINICAL · NOSE & SINUSES

Sinonasal tumours

Tumours of the nose and sinuses hide behind the symptoms of ordinary sinusitis until they break out of their bony box into the cheek, palate, orbit or skull base. The warning is a one-sided story: blockage, bloodstained discharge, numbness or a loose tooth on one side in a middle-aged or older adult. Every one-sided nasal mass is sent for histology.

AT A GLANCE

- One-sided symptoms in an adult mean scope and biopsy.
- Staging with CT and MRI.
- Surgery then radiotherapy for most.
- Numbness and eye signs are alarming.
- SNUC responds to induction chemotherapy; endoscopic resection matches open results with clear margins.

1 CHAPTER ONE Overview

Silent until they escape.

1.1 The range

SINONASAL TUMOURS

- BENIGN EPITHELIAL **Papillomas** squamous, inverted, oncocytic, exophytic
- BENIGN OTHER **Haemangioma, schwannoma, meningioma** pleomorphic adenoma, chondroma, angiofibroma
- BENIGN BONY **Osteoma, fibrous dysplasia** ossifying fibroma, ameloblastoma
- MALIGNANT EPITHELIAL **Squamous cell carcinoma** adenocarcinoma, adenoid cystic carcinoma, undifferentiated carcinoma
- MALIGNANT OTHER **Melanoma, olfactory neuroblastoma** lymphoma, plasmacytoma, sarcomas, haemangiopericytoma

1.2 Epidemiology

Sinonasal cancer is rare (well under 1% of all cancers). The maxillary sinus is the commonest site, then the ethmoids; frontal and sphenoid primaries are rare.

1.3 Risk factors

- **Hardwood dust** (furniture workers): adenocarcinoma of the ethmoids and upper nose.
- **Nickel refining:** squamous and anaplastic carcinoma.
- **Leather work, chromium, mustard gas, formaldehyde.**
- **Snuff** rich in nickel and chromium (Bantu populations).
- **Smoking** and HPV for some squamous carcinomas.

1.4 Histology

More than 80% of malignant tumours are squamous cell carcinoma; the rest are adenocarcinoma, adenoid cystic carcinoma, melanoma, olfactory neuroblastoma, lymphoma and sarcomas.

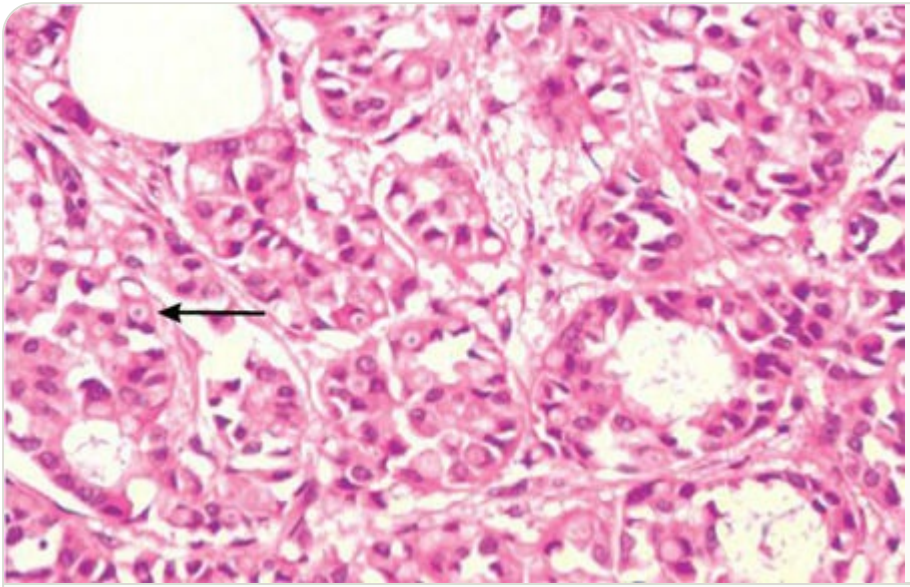


FIGURE 1 Adenocarcinoma with gland formation and intracytoplasmic mucin.
Dhingra, fig. 40.4

2 CHAPTER TWO
Benign tumours of the nasal cavity

And the one that isn't quite benign.

2.1 Squamous papilloma

A warty lesion of the vestibule or anterior septum; excised.



FIGURE 2 Squamous papilloma of the left nasal vestibule.
Dhingra, fig. 39.1

2.2 Inverted papilloma

2.2.1 What it is

A Schneiderian papilloma in which the epithelium grows inwards into the stroma. Typically in men of 40 to 70, almost always one-sided, arising from the lateral wall near the middle meatus and extending into the maxillary and ethmoid sinuses. HPV is found in some.

2.2.2 Three warnings

- It destroys bone by pressure.
- It recurs if incompletely removed.

- It is associated with squamous cell carcinoma in about 10%.

2.2.3 Clinical picture

One-sided obstruction, discharge and bleeding, with a firm, greyish, irregular mass that can masquerade as a simple polyp.

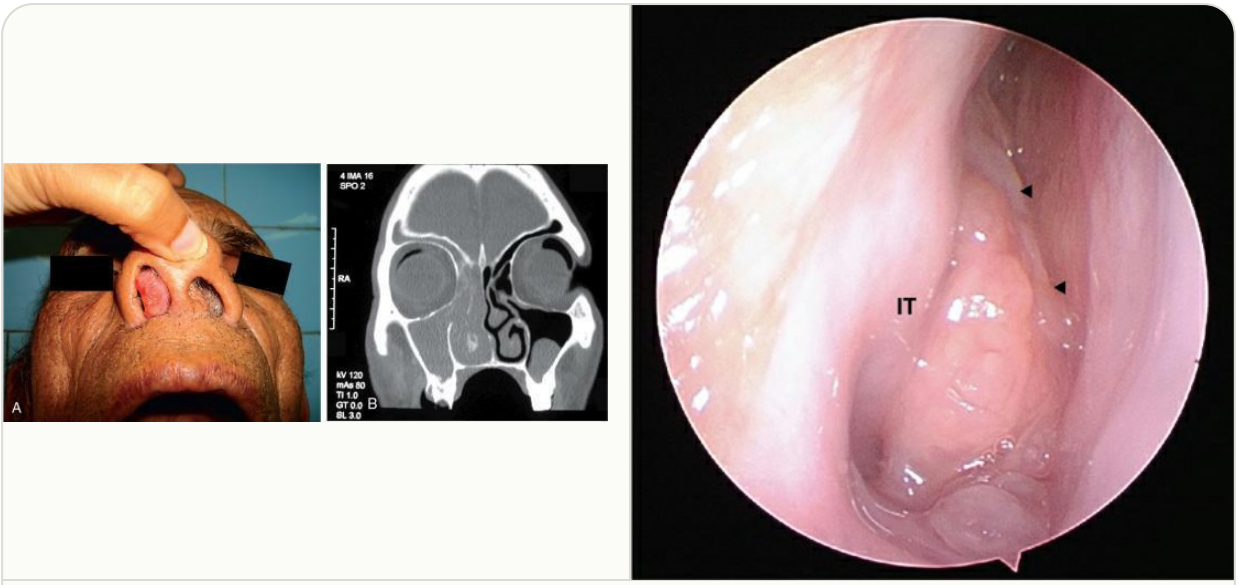


FIGURE 3 First, inverted papilloma in a 79-year-old man, with its CT. Second, an inverted papilloma in the right nose looking like a simple polyp.

Dhingra, figs 39.2 and 39.3

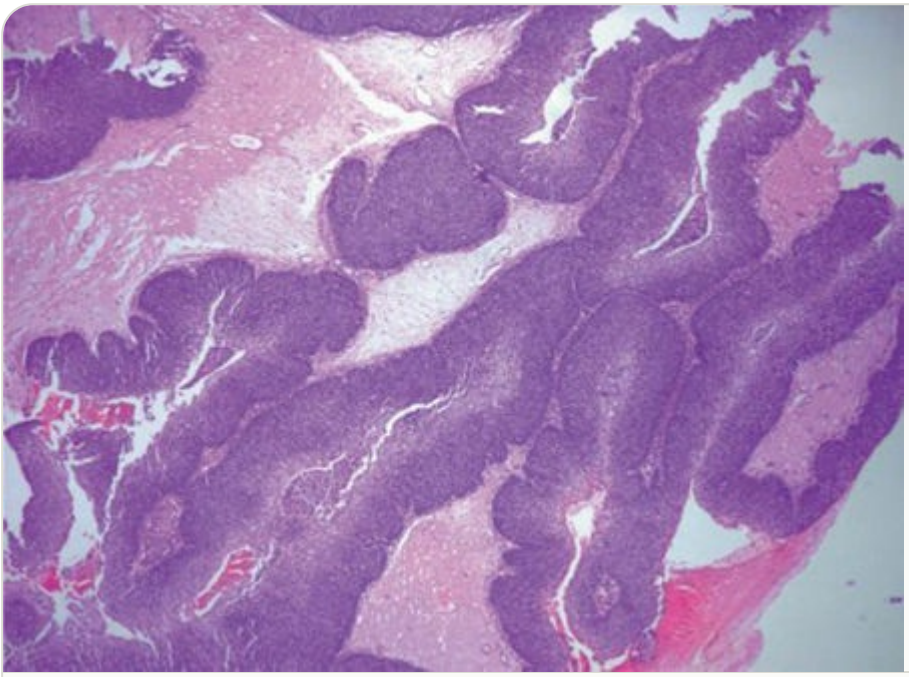


FIGURE 4 Histology of a Schneiderian papilloma: ribbons of thickened epithelium growing into the stroma.

Dhingra, fig. 39.4

2.2.4 Imaging

CT shows a one-sided mass and focal hyperostosis at the site of attachment; MRI shows a *convoluted cerebriform* pattern.

2.2.5 Treatment

Complete removal including the mucoperiosteum and drilling of the bone at the site of attachment, now usually endoscopic (endoscopic medial maxillectomy) and formerly by lateral rhinotomy or midfacial degloving. Long-term endoscopic follow-up.

2.3 Vascular and other benign tumours

- **Haemangioma** (lobular capillary haemangioma, *bleeding polyp of the septum*): a red, friable mass on the anterior septum, often in pregnancy; excised with its base.
- **Schwannoma, meningioma, chondroma, pleomorphic adenoma** of the septum.
- **Angiofibroma** (see its own topic).
- **Encephalocele, glioma and dermoid** (see the congenital nasal masses topic).



FIGURE 5 A bleeding polyp arising from the right side of the septum.
Dhingra, fig. 39.5

3 CHAPTER THREE
Benign bony tumours

Usually found by chance.

3.1 Osteoma

The commonest benign sinus tumour, usually in the frontal sinus, then ethmoid; often an incidental finding on CT. Symptoms arise when it blocks drainage (mucocele, sinusitis), bulges into the orbit or erodes the skull base. Multiple osteomas suggest *Gardner syndrome* (with colonic polyposis). Removed endoscopically or through an external approach when symptomatic.

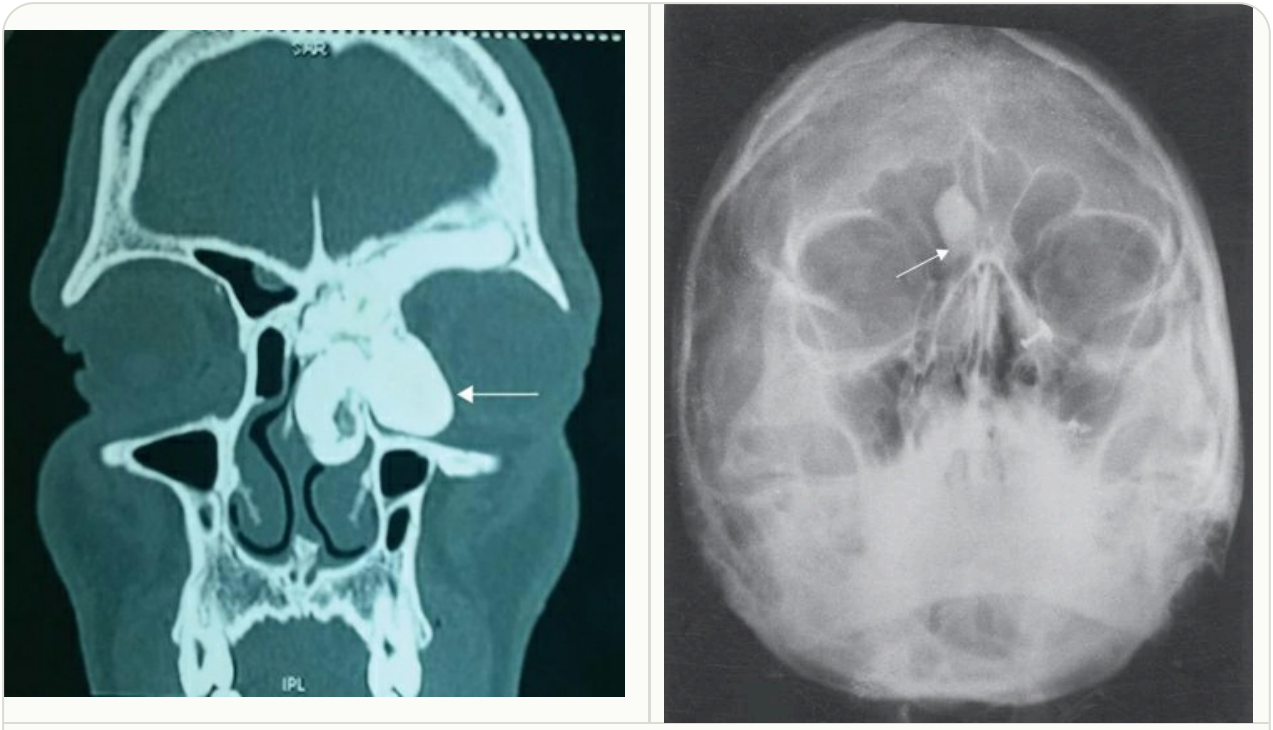


FIGURE 6 First, a frontoethmoidal osteoma invading the orbit. Second, an osteoma of the right frontal sinus.

Dhingra, figs 40.1 and 40.2

3.2 Fibrous dysplasia

Bone replaced by fibrous tissue and woven bone; *monostotic*, *polyostotic*, or with skin pigmentation and endocrine disorders (*McCune-Albright syndrome*). A painless swelling of the maxilla in a child or adolescent, with a *ground-glass* appearance on CT.

Growth usually slows after puberty; conservative recontouring for deformity or compression.



FIGURE 7 Fibrous dysplasia of the maxilla in a 13-year-old girl.
Dhingra, fig. 40.3

3.3 Ossifying fibroma

A well-defined, expansile lesion with a bony rim, more aggressive in its juvenile form; excised completely.

3.4 Ameloblastoma

An odontogenic tumour of the jaw, commonest in the mandible, that may invade the maxillary sinus; locally aggressive and recurrent unless widely excised.

IMAGING IN THE APP 4 cases

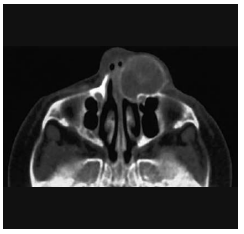
Inverted papilloma, osteoma, fibrous dysplasia and myxoma.



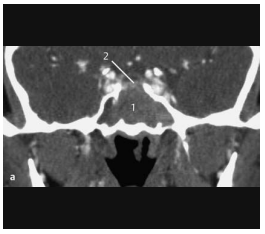
Osteoma
Bone disease · 3 images



Fibrous Dysplasia of the Sinuses (Monostotic and Polyostotic)
Bone disease · 5 images



Myxoma
Tumours · 1 image



Inverted Papilloma
Tumours · 12 images

4 CHAPTER FOUR Carcinoma of the maxillary sinus

The classic sinonasal cancer.

4.1 Clinical picture

Usually men of 40 to 60. Early symptoms are easily mistaken for sinusitis: nasal stuffiness, bloodstained discharge, facial pain or numbness, and epiphora. Late features depend on the direction of spread.

SPREAD OF MAXILLARY CARCINOMA

DIRECTION	STRUCTURES	FEATURES
Medial	Nasal cavity, ethmoids	Obstruction, discharge, epistaxis
Anterior	Cheek	Swelling, then skin invasion
Inferior	Alveolus, palate	Toothache, loose teeth, ill-fitting dentures, palatal swelling
Superior	Orbit	Proptosis, diplopia, pain, epiphora
Posterior	Pterygoid plates and muscles, infratemporal fossa	Trismus, spread to nasopharynx and skull base
Intracranial	Through the ethmoids, cribriform plate or foramen lacerum	Neurological signs



FIGURE 8 Carcinoma of the left maxilla in a 75-year-old man: (A) swelling and redness of the cheek, (B) expansion of the upper alveolus with an intraoral growth, (C) CT showing destruction of the maxilla and orbital floor.

Dhingra, fig. 40.6

4.2 Lymph node and distant spread

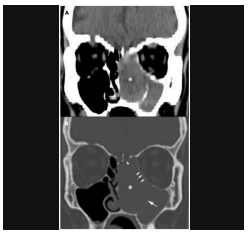
Nodes are uncommon and late (submandibular and upper jugular); the first-echelon retropharyngeal nodes cannot be felt. Distant metastases go to the lungs and bone.

4.3 Investigations

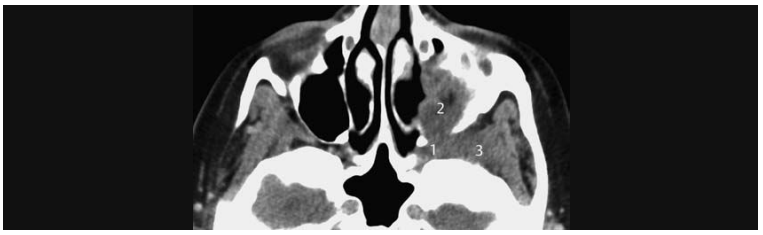
- **CT** in three planes: bone destruction and staging.
- **MRI:** separates tumour from retained secretions, and shows orbital, perineural and intracranial spread.
- **Biopsy:** endoscopic, or from the mouth or nose when the tumour presents there; the Caldwell-Luc route is no longer preferred.
- **Staging:** chest CT or PET-CT.

IMAGING IN THE APP 2 cases

Sinonasal malignancy and spread into the pterygopalatine and infratemporal fossae.



Sinonasal tumours
Tumours · 10 images



Malignancy of the Pterygopalatine and Infratemporal Fossa
Tumours · 2 images

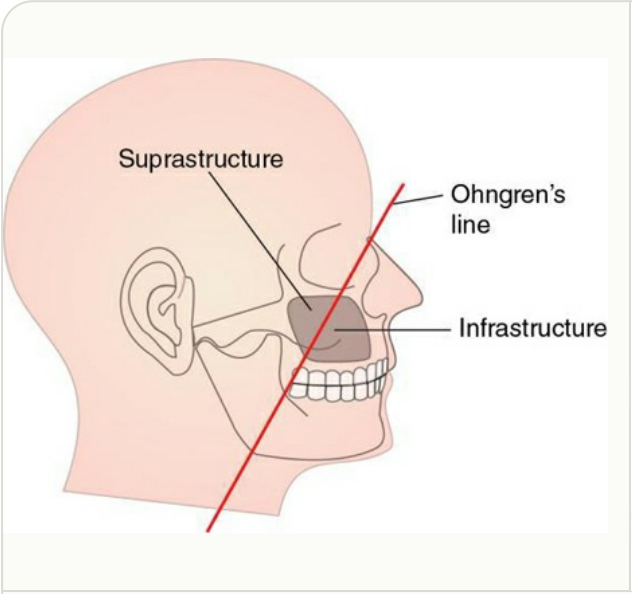
4.4 Classifications

4.4.1 Ohngren's line

An imaginary line from the medial canthus to the angle of the mandible. Tumours above and behind it (suprastructure) have a worse prognosis than those below and in front of it.

4.4.2 Lederman's classification

Two horizontal lines of Sebileau, through the floors of the orbits and the floors of the antra, divide the region into *suprastructure* (ethmoid, sphenoid, frontal sinuses and olfactory area), *mesostructure* (maxillary sinus and respiratory nose) and *infrastructure* (alveolus).



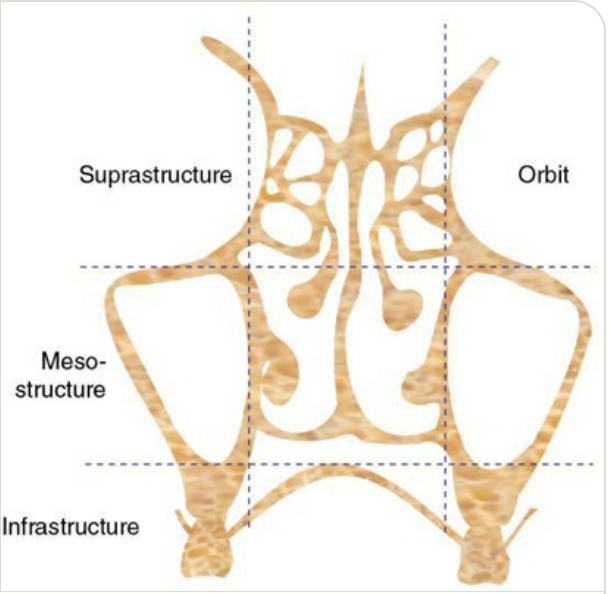


FIGURE 9 First, Ohngren's line. Second, Lederman's classification.
Dhingra, figs 40.7 and 40.8

4.4.3 AJCC TNM for the maxillary sinus

T STAGING OF MAXILLARY SINUS CARCINOMA

T	EXTENT
T1	Limited to the mucosa, no bone erosion
T2	Bone erosion including the hard palate or middle meatus, but not the posterior wall or pterygoid plates
T3	Posterior wall, subcutaneous tissue, floor or medial wall of the orbit, pterygoid fossa, ethmoids
T4a	Anterior orbit, cheek skin, pterygoid plates, infratemporal fossa, cribriform plate, sphenoid or frontal sinus
T4b	Orbital apex, dura, brain, middle cranial fossa, cranial nerves other than V2, nasopharynx, clivus

4.5 Treatment

4.5.1 Surgery

Resection with clear margins, followed by radiotherapy for most stages:

- **Medial maxillectomy** (endoscopic or open) for medial tumours.
- **Total maxillectomy** through a *Weber-Fergusson* incision (lateral rhinotomy with lip split, extended below the eye), with **orbital exenteration** when the orbit is invaded beyond the periosteum.
- **Craniofacial resection** when the cribriform plate or anterior skull base is involved.
- Endoscopic resection is increasingly used for selected tumours.



FIGURE 10 First, the Weber-Fergusson incision. Second, maxillectomy with orbital exenteration, before and after rehabilitation.
Dhingra, figs 40.10 and 40.9

4.5.2 Reconstruction and rehabilitation

A dental obturator closes the palatal defect, or a free flap reconstructs it; orbital prostheses after exenteration.

4.5.3 Radiotherapy and chemotherapy

Postoperative radiotherapy for most; primary chemoradiotherapy for unresectable tumours; neoadjuvant chemotherapy in selected cases.

4.5.4 Neck

Neck dissection or irradiation for nodal disease, and elective treatment in some advanced tumours.

OPERATIVE GUIDES IN THEATRE 4 operations

Medial Maxillectomy
Medial maxillectomy

Complete Maxillectomy
Complete maxillectomy

Transcranial and Endonasal Approaches to the Anterior Cranial Base
Anterior skull base approaches

Orbital Exenteration
Exenteration

4.6 Prognosis

Five-year survival is around a third to a half overall, worse with orbital apex, skull base or intracranial spread.

5 CHAPTER FIVE
Tumours of the ethmoid, frontal and sphenoid

Rarer sites.

5.1 Ethmoid

Often adenocarcinoma in woodworkers. Nasal obstruction, bloodstained discharge, a mass in the medial canthus, proptosis with lateral displacement of the eye, and early skull base spread. Craniofacial or endoscopic resection with radiotherapy.

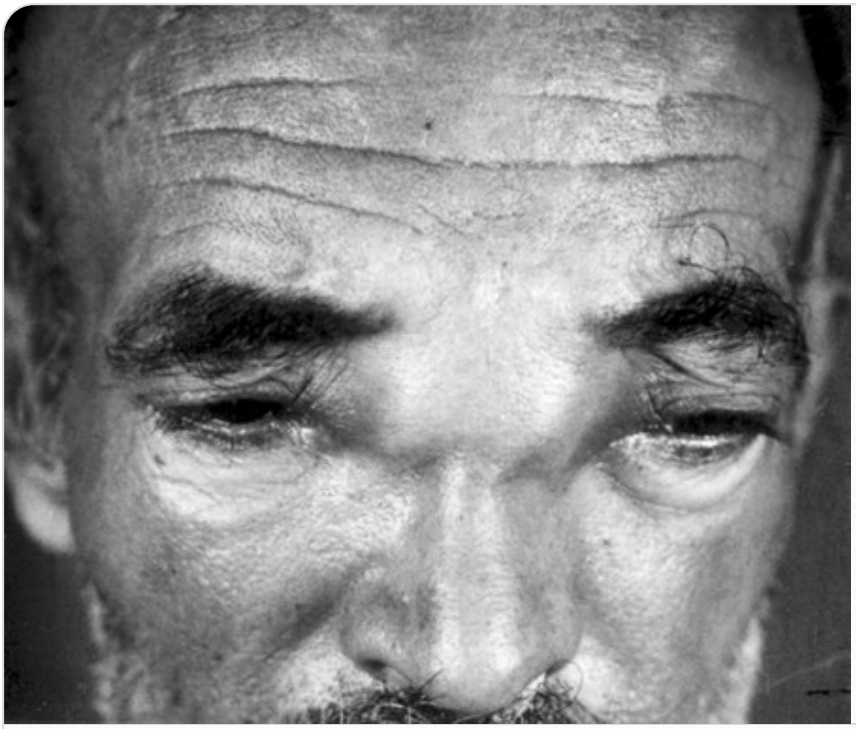


FIGURE 11 Carcinoma of the ethmoid.
Dhingra, fig. 40.11

5.2 Frontal

Very rare: forehead swelling, frontal pain, displacement of the eye downwards, and early intracranial spread.

5.3 Sphenoid

Usually secondary or from the nasopharynx; presents with cranial nerve palsies (second to sixth) and headache.

6 CHAPTER SIX Other malignant tumours

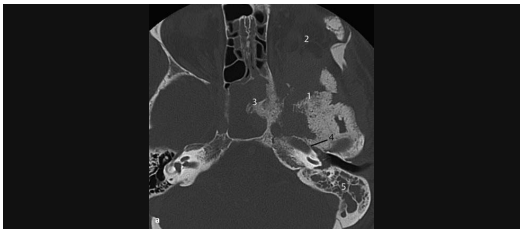
Each with its clue.

6.1 Olfactory neuroblastoma

Esthesioneuroblastoma, from the olfactory epithelium at the roof of the nose, at any age with peaks in the second and sixth decades. One-sided obstruction, epistaxis and loss of smell, with a mass high in the nose that may extend through the cribriform plate (a *dumb-bell* shape). Staged by the Kadish system (A nasal cavity, B sinuses, C beyond); treated by craniofacial or endoscopic resection with radiotherapy. Late neck metastases occur.

IMAGING IN THE APP 1 case

A meningioma destroying the skull base, a differential of olfactory neuroblastoma.



Meningioma with Skull Base Destruction

Tumours · 2 images

6.2 Mucosal melanoma

Older adults, on the septum or lateral wall; pigmented or amelanotic, bleeding; spreads by blood and lymph and has a poor prognosis; wide excision and radiotherapy.

6.3 Lymphoma, plasmacytoma and sarcomas

NK/T-cell lymphoma (see the granulomatous disease topic), extramedullary plasmacytoma, and rhabdomyosarcoma in children.



FIGURE 12 Rhabdomyosarcoma of the nose in a two-and-a-half-year-old boy.
Dhingra, fig. 39.7

THE NEW CLASSIFICATION AND TREATMENT TRENDS (OXFORD TEXTBOOK)

The fourth WHO classification reorganised sinonasal tumours and added new entities, so histology now carries more weight than ever. **Sinonasal undifferentiated carcinoma (SNUC)** is aggressive, often presents with intracranial and nodal spread, and responds well to **induction chemotherapy** followed by chemoradiotherapy, with surgery when the response is poor. **NUT carcinoma**, defined by a *NUTM1* rearrangement, affects young adults and has a very poor prognosis. **Neuroendocrine carcinoma** (small and large cell types) is treated with combined chemotherapy and radiotherapy. **Intestinal-type adenocarcinoma**, linked to hardwood dust, looks like bowel cancer under the microscope but is not a metastasis; it is often multifocal, so wide removal of the sinus mucosa is advised. Low-grade non-intestinal adenocarcinoma may need surgery alone; high-grade disease needs radiotherapy too.

SCC arising in an inverted papilloma behaves like de novo SCC; about 5% of inverted papillomas harbour dysplasia or carcinoma, so the whole specimen must be examined.

Endoscopic resection now achieves oncological outcomes equivalent to open surgery for most tumours, with less morbidity, provided clear margins can be obtained; advanced tumours involving the orbit or skull base may first have induction chemotherapy, with the response guiding radiotherapy or surgery. Nodal metastases at presentation suggest a high-grade tumour such as SCC or SNUC, and up to 20% of olfactory neuroblastomas spread to the neck, so long-term neck surveillance is needed. Somatostatin-receptor PET (DOTATATE) helps stage and follow olfactory neuroblastoma and neuroendocrine tumours. The Kadish system for olfactory neuroblastoma discriminates poorly, and the Dulguerov TNM-based system is often preferred.

CHAPTER SEVEN

7 For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Occupational risks:** wood dust and adenocarcinoma, nickel and squamous carcinoma.
- **Inverted papilloma:** site, three warnings, imaging and treatment.
- **Maxillary carcinoma:** symptoms by direction of spread.
- **Ohngren's line and Lederman's classification.**
- **T staging,** and maxillectomy with or without orbital exenteration.
- **Olfactory neuroblastoma** and Kadish staging.

7.2 Common pitfalls

COMMON PITFALLS

- Treating one-sided sinus symptoms with antibiotics for months.

- Removing a one-sided polyp without sending it for histology.
- Calling numbness of the cheek sinusitis.

Written for this app from Oxford Textbook of Otolaryngology ch. 39, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 9, Dhingra ch. 39, 40 and K. J. Lee ch. 32. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · TUMOURS

Inverted Papilloma

IMAGING FINDINGS

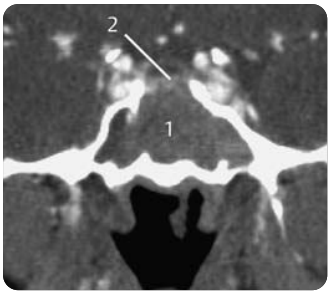
- Unilateral mass on the lateral nasal wall near the middle meatus
- Focal hyperostosis at the attachment
- Convoluted cerebriform pattern on T2 and post-contrast MRI

TOP DIFFERENTIALS

- Antrochoanal or solitary polyp
- Squamous cell carcinoma (in up to about 10%)
- Juvenile angiofibroma

CHECKLIST

- MRI separates tumour from secretions
- Bone destruction raises concern for carcinoma



Patient operated before because of nasal polyps. Histologic evaluation revealed inverted papilloma. Follow-up scans are shown.…



Axial bone CT shows a mass in the nasal cavity along the lateral wall near the middle meatus. The maxillary sinus is opacified, but it is difficult to…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 744.

PARANASAL SINUSES · TUMOURS

Sinonasal tumours

IMAGING FINDINGS

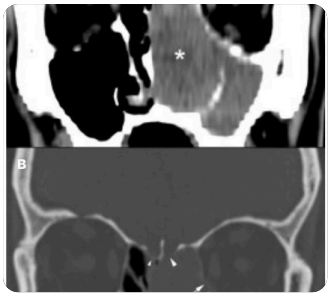
- Unilateral solid mass with bone destruction
- Intermediate T2, solid enhancement, distinct from bright secretions
- Spread to orbit, skull base, pterygopalatine fossa, cheek

TOP DIFFERENTIALS

- Inverted papilloma
- Lymphoma
- Olfactory neuroblastoma, adenocarcinoma, melanoma

CHECKLIST

- Periorbita and dura
- Perineural spread along V2 and the vidian nerve; nodes



Coronal reformatted images from a noncontrast CT in soft tissue window (a) and bone window settings (b). The mass (asterisk) causes resorption of…



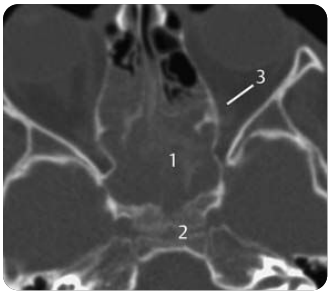
Axial CECT shows the typical location and appearance of an antral SCCa. There is extension into the premaxillary soft tissues anteriorly an…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 754.

NOSE & NASAL CAVITY · TUMOURS

Olfactory Neuroblastoma (Olfactory neuroblastoma (esthesioneuroblastoma))

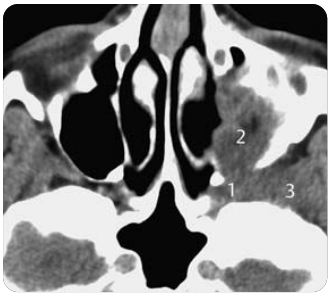
- Olfactory neuroblastoma (esthesioneuroblastoma) is an uncommon tumor of neural crest origin that arises from the olfactory mucosa in the superior nasal cavity. These polypoid tumors typically start unilaterally and may bleed profusely on biopsy. Neglected cases appear as a bilateral mass in the anterior cranial fossa. In these larger tumors, which usually have intracranial extension, peripheral tumoral cysts can occur at the margins of the intracranial part of the mass. These cysts have their broadest base on the tumor and, when present, are highly suggestive of the (imaging) diagnosis of olfactory neuroblastoma. Beware of encroachment of cranial nerves or intracranial vessels, which may be important in the estimation of morbidity during removal.



CT, axial bone window. There is complete bilateral opacification of the ethmoid and sphenoid sinuses (1). The anterior cortex of the clivus (2) seem…

INFRATEMPORAL FOSSA & TMJ · TUMOURS

Malignancy of the Pterygopalatine and Infratemporal Fossa

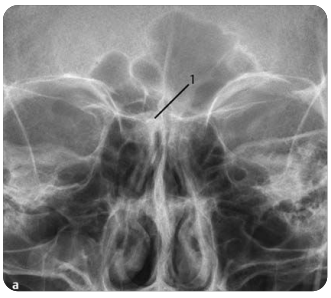


CT, axial with intravenous contrast enhancement. In the pterygopalatine fossa, a lesion is seen (1) with bony destruction and invasion into the…

PARANASAL SINUSES · BONE DISEASE

Osteoma

- Large osteomas may cause functional problems due to growth and encroachment of surrounding structures (see Osteoma, Gardner Syndrome, and Complications). For this reason, and also in small osteomas, a wait-and-scan policy might be advisable.

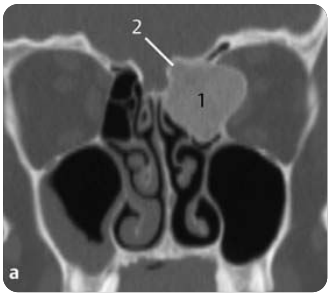


Plain radiograph, Caldwell view. Stasis of secretions in the frontal recess or frontal sinus is difficult to determine on conventional radiographs, as is th…

PARANASAL SINUSES · BONE DISEASE

Fibrous Dysplasia of the Sinuses (Monostotic and Polyostotic)

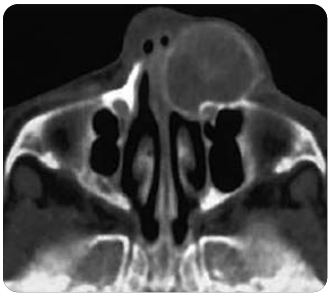
-
- Expansion of bone with a ground glass appearance and intact cortex reflects the pathophysiology of this disease (i.e., replacement of normal medullary bone by woven fibro-osseous tissue).
- The lesion may be misinterpreted as a mucocele or an osteoma, depending on the CT window setting.
- An ossifying fibroma may demonstrate central bony densities in later stages.
- Secondary sinusitis and vascular or neurological deficits may develop with progression of the lesions due to encroachment of sinus ostia, fissures, skull-base foramina, and/or the optic canal. Esthetic and functional problems due to lesions in the mandible and maxilla are not uncommon.
- Expansion of bone with a ground glass appearance and intact cortex reflects the pathophysiology of this disease (i.e., replacement of normal medullary bone by woven fibro-osseous tissue). –
- A separate variant of polyostotic fibrous dysplasia is seen in McCune–Albright syndrome, with severe progressive lesions in multiple locations (also seen more often in females, and with cutaneous pigmentations and precocious puberty).



CT, coronal, bone-tissue setting. A solitary lesion (1) is located in the left ethmoid and is obstructing the frontal recess with expansion into the left…

PARANASAL SINUSES · TUMOURS

Myxoma



CT, axial. Bony expansion and remodeling observed on CT. Pathologic examination after complete removal revealed a myxoma.…

NOSE & NASAL CAVITY · CONGENITAL

Meningoencephalocele

IMAGING FINDINGS

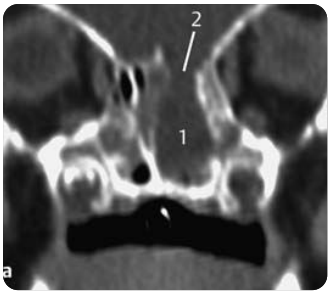
- Bony defect in the cribriform plate, lateral lamella, ethmoid roof or sphenoid
- Soft tissue or fluid through the defect
- MRI: CSF, meninges or brain in continuity with the intracranial contents

TOP DIFFERENTIALS

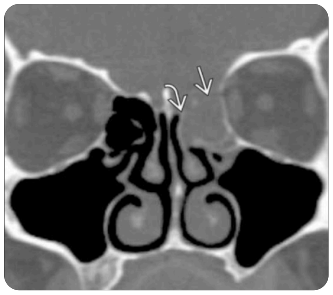
- Skull base defect without leak
- Sinus opacification from mucus

CHECKLIST

- Lateral sphenoid recess and multiple defects
- Signs of raised intracranial pressure: empty sella



CT, coronal. The left nasal cavity is filled with a soft-tissue mass (1) that seems to be connected to brain through a defect in the roof of the…



Coronal bone CT shows large bony defect in left ethmoid roof, lateral to insertion of middle turbinate. Because there is complete opacification of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 920.